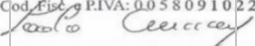


UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

Client	PBDiM - Energopol
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

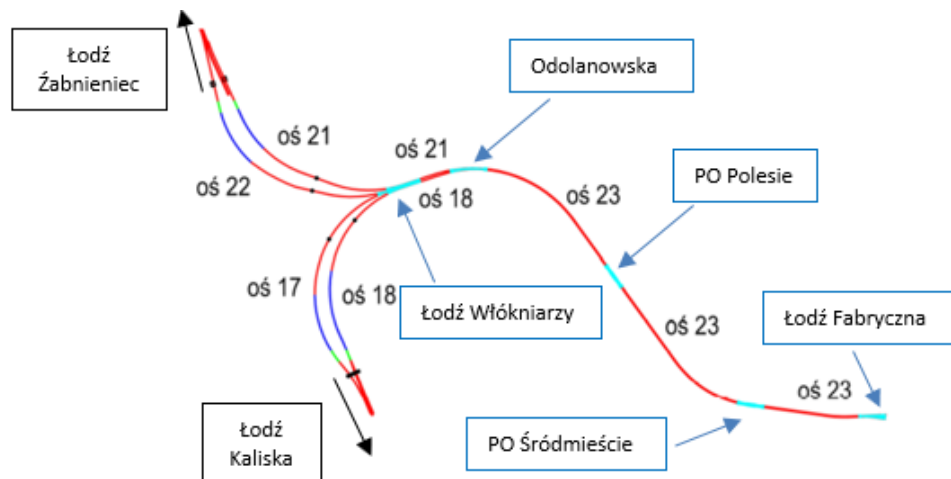


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 05/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+637.31 after the erection of ring n. 264; which corresponds to a total excavation length of 329.16 m from the starting chainage.

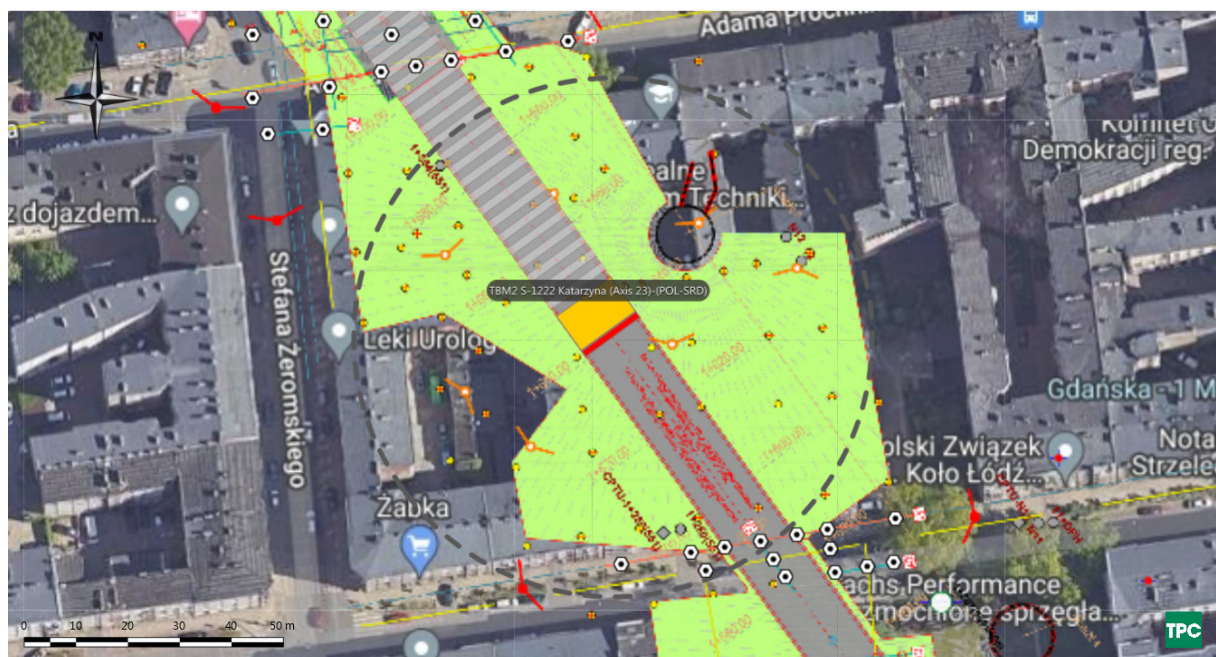


Figure 1-2 Plan view of TBM Position on 06/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.6 bar, the minimum attention value. It was noted that a drop in pressures below minimum design pressures occurred during the excavation of ring 259, to reduce counter forces to the shield and to advance. The values are then recovered within the attention values.

Figure 2-1 shows EPB sensors.

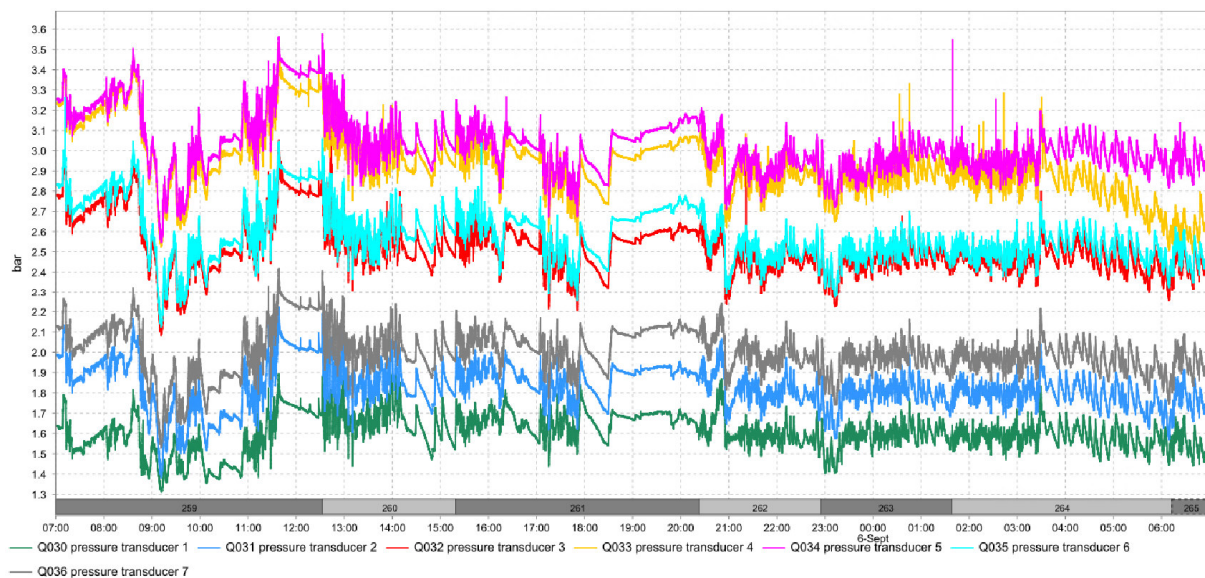


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.22 bar indicating that the bulk chamber is correctly filled with material. It was noted that a drop in delta pressure occurred during the excavation of ring 259.

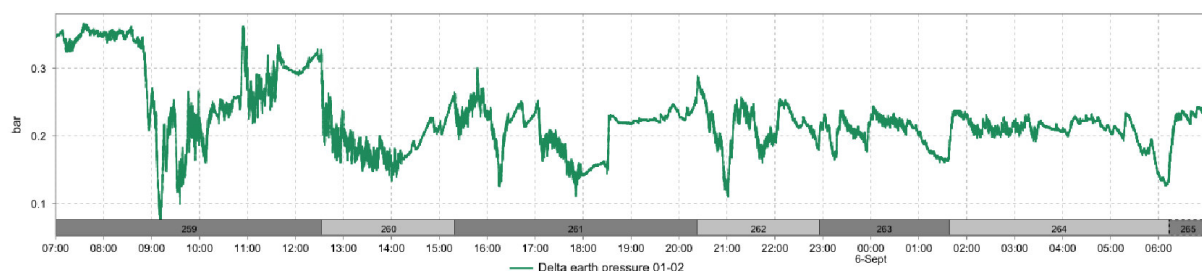


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 shows lower values than n. 1. Observed values for rings 259 and 260 are not representative. The belt scale value is a raw indication of the excavated material quantity.

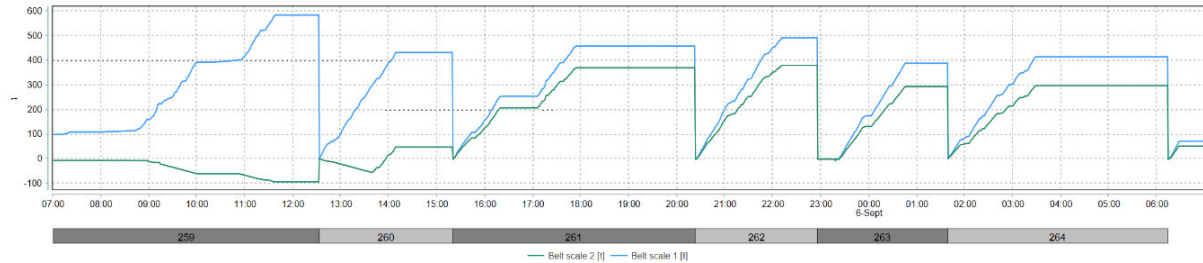


Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
259	586	0
260	434	50
261	461	372
262	493	382
263	390	296
264	417	299

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 8.5 MNm.

During advancement, average advance rate is of 16 mm/min.

During excavation, thrust force shows an average value of 64.4 MN and a maximum of 89.5 MN, reached during the excavation of ring 259.

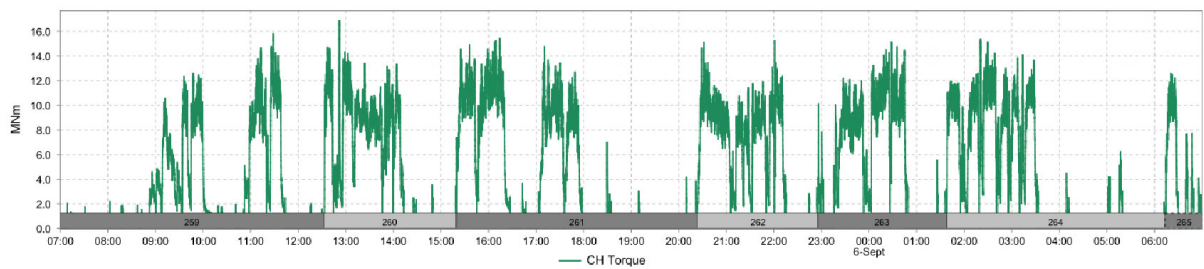


Figure 3-1 CH Torque

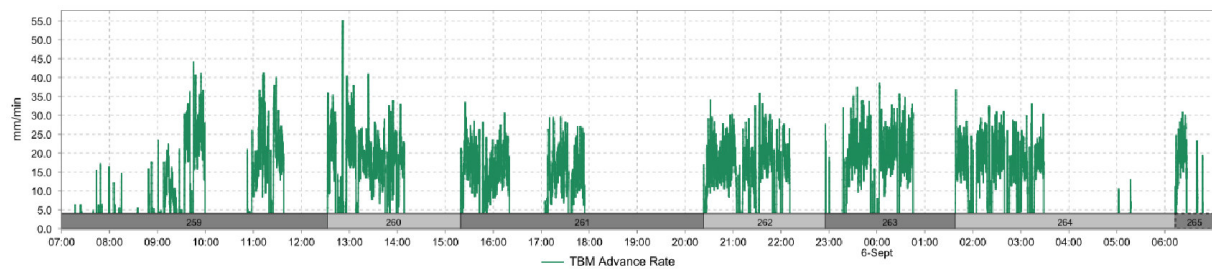


Figure 3-2 Advance rate

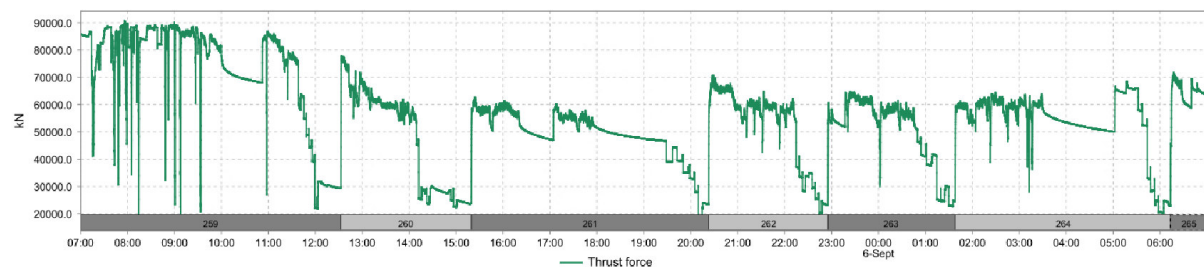


Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume closer to the minimum theoretical annular volume $\pm 10\%$ of 13.54 m^3 . Ratio B/A is at 7%.

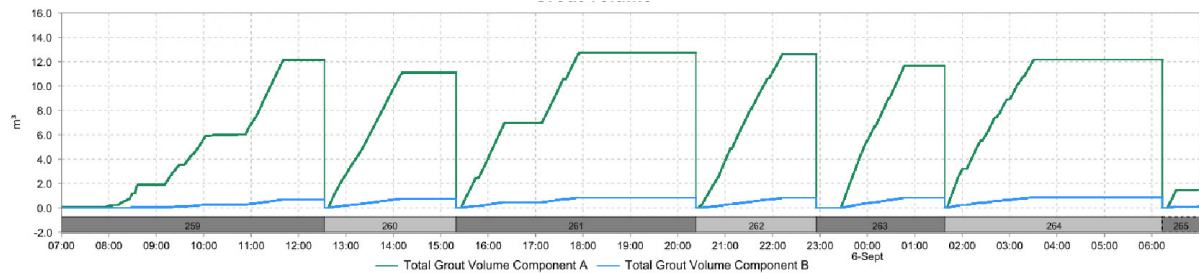


Figure 4-1 Grout volumes

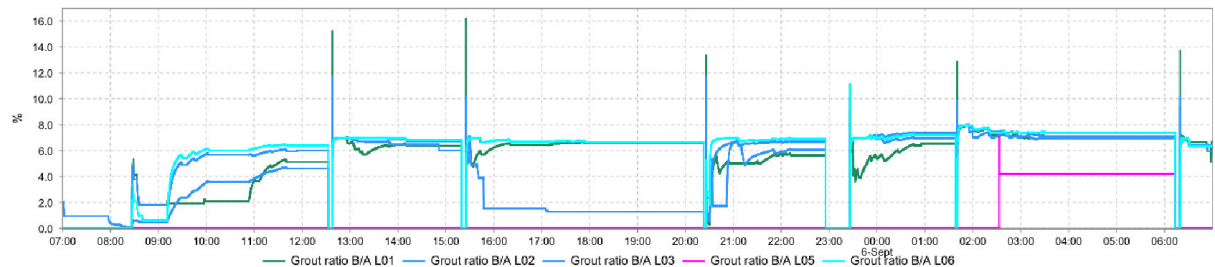


Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
259	12.1	0.7	5
260	11.1	0.7	7
261	12.7	0.8	7
262	12.6	0.8	6
263	11.7	0.8	7
264	12.2	0.9	7

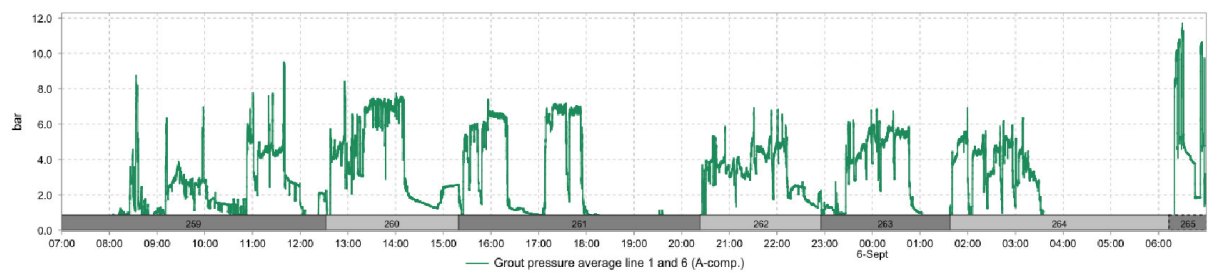


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -6.86 mm on the front edge of the shield and the horizontal deviation of about -8.23 mm (Figure 5-1). Maximum radial deviation is of 14.5 mm.

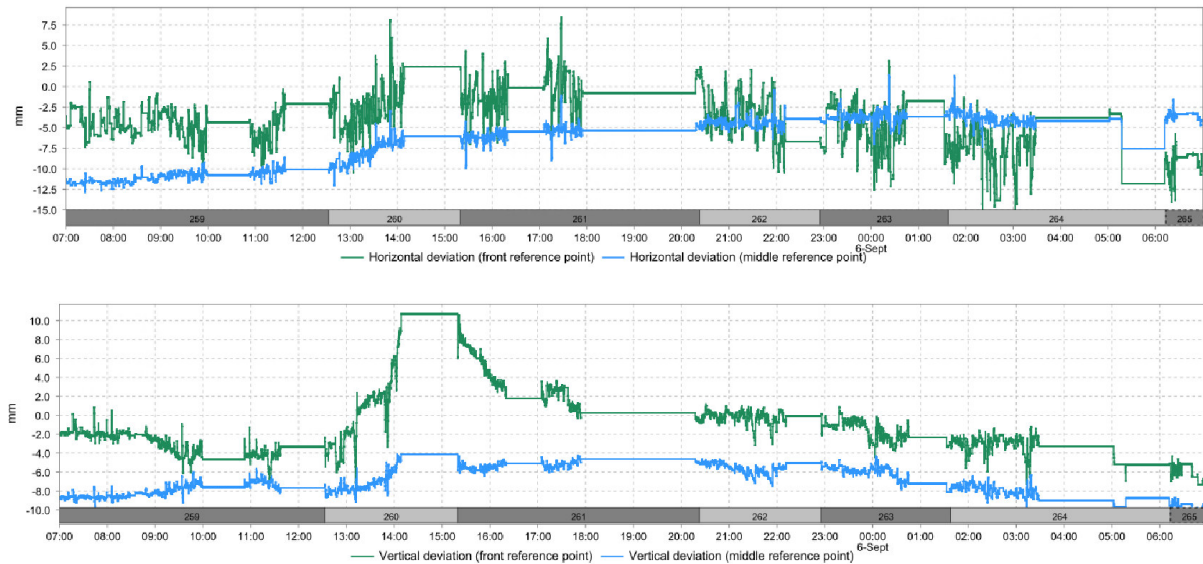


Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.



Figure 6-1 Monitoring points

On PR0440, monitoring points show a small uplift trend for point PR0440_03R and a downward trend for point PR0440_05R.

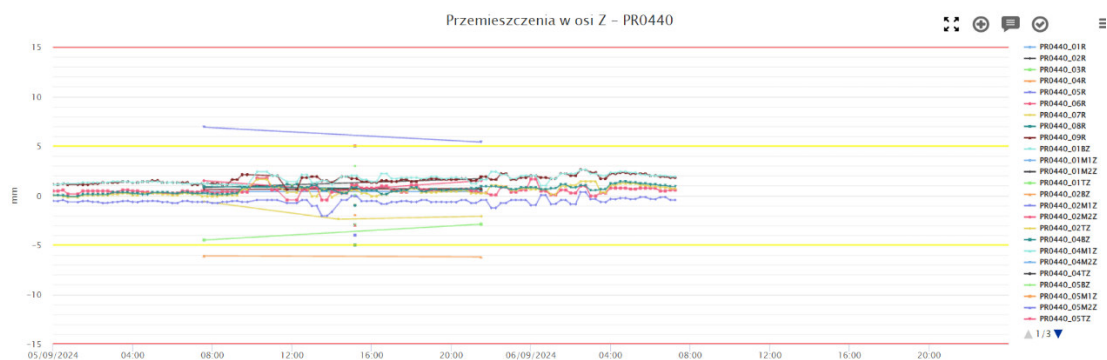


Figure 6-2 Settlement of building PR0440

On PR0460, monitoring points are stable.

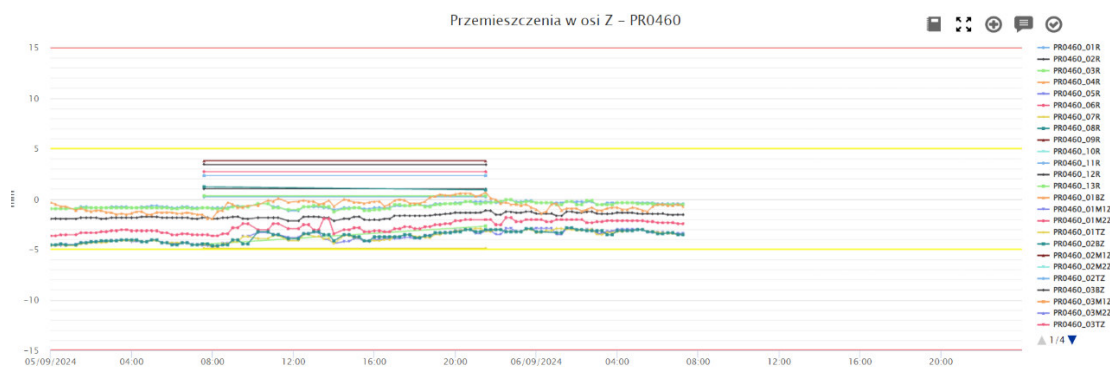


Figure 6-3 Settlement of building PR0460

On MA0190, monitoring points are stable.

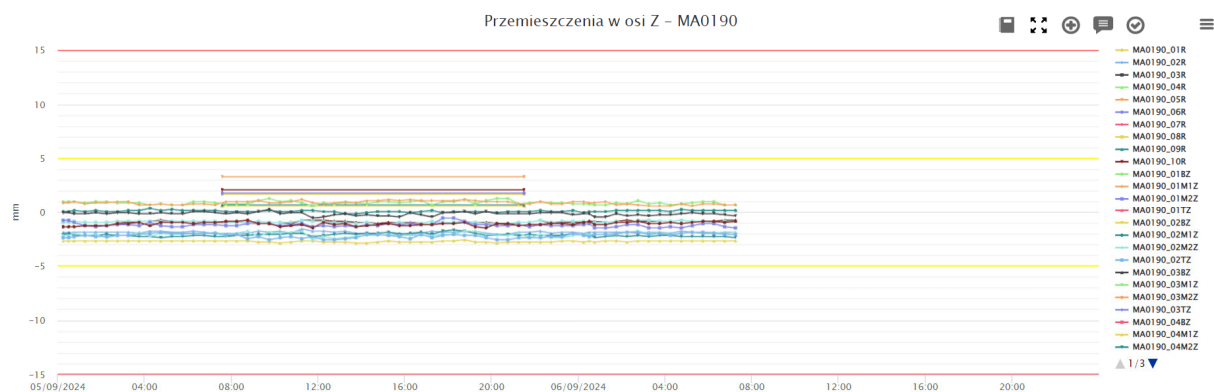


Figure 6-4 Settlement of building MA0190

On MA0210, monitoring points are stable.

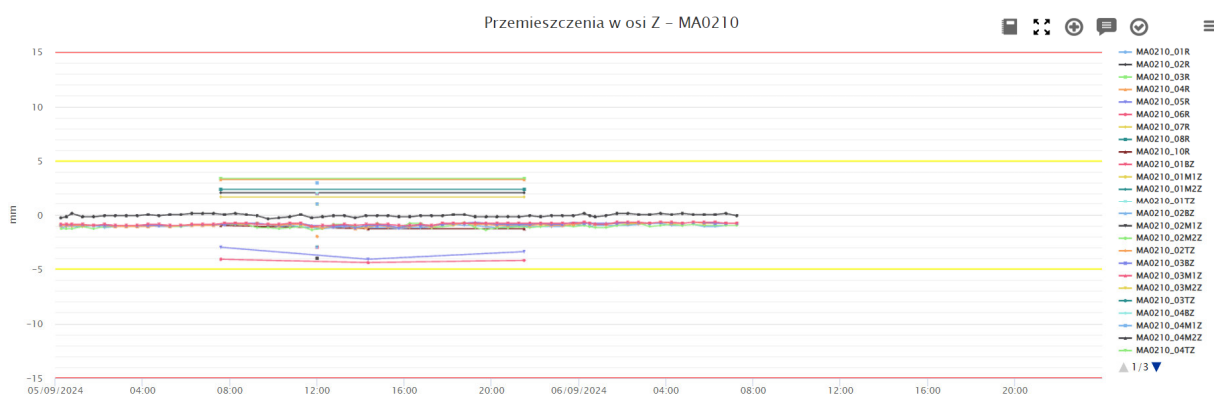


Figure 6-5 Settlement of building MA0210

On MA0230, monitoring points show a small uplift trend for point MA0230_04R

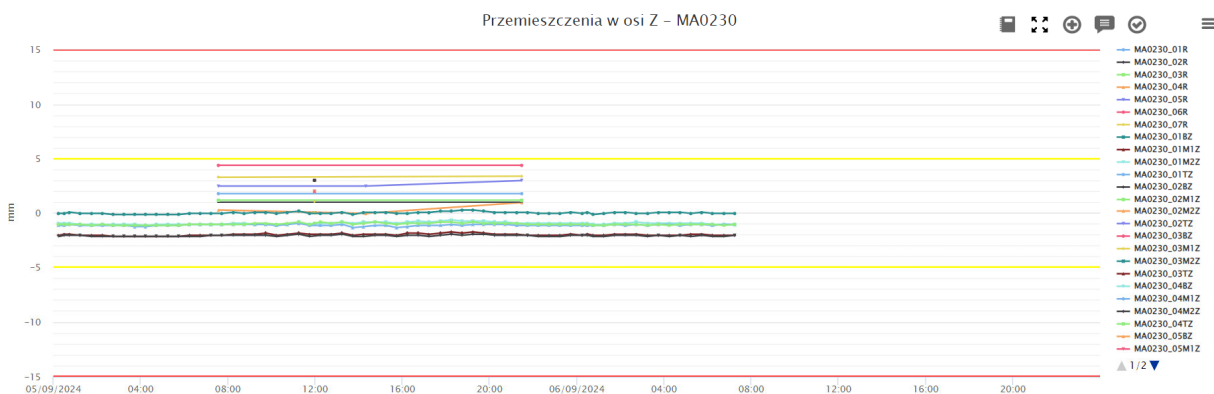


Figure 6-6 Settlement of building MA0230

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 05/09/2024 of Soletanche, compensation grouting has been performed on the 05/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 30225.93 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

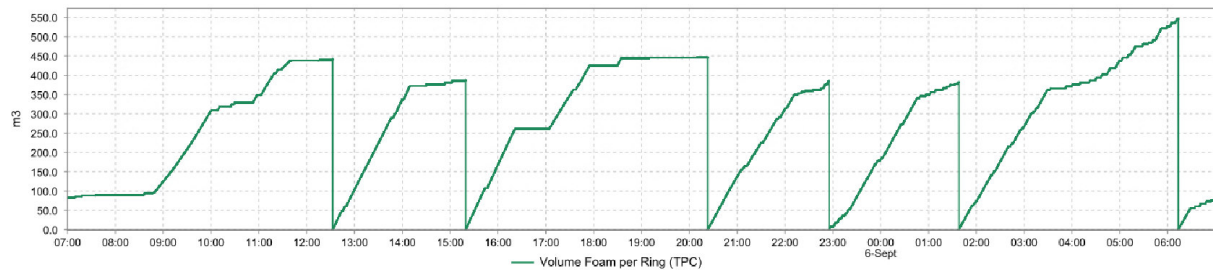


Figure 8-1 Injected foam volume

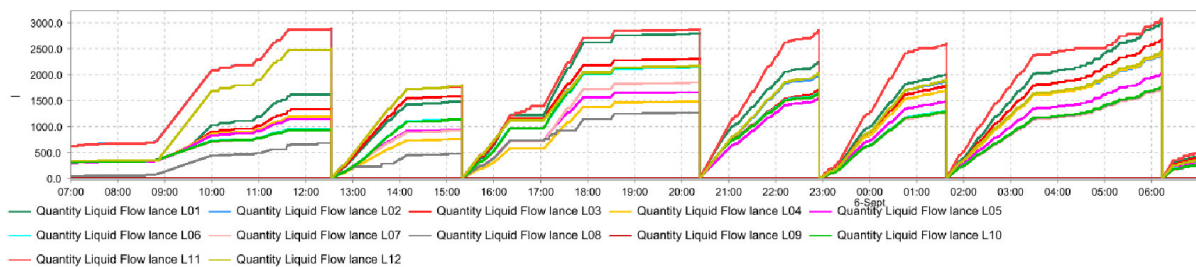


Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m³)	C (%)	FIR (%)
259	17	1.9	207
260	13.9	2	182
261	22.9	2	211
262	20.5	2	182
263	18.5	2	181
264	25	1.9	258